1. Setting up a Mail Server

- **Description**: Set up a mail server using Postfix or Sendmail. Implement SMTP for sending emails and either IMAP/POP for receiving. Secure the server with TLS/SSL and set up a basic spam filtering system. Test email delivery within a local network and to external servers.

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2. Setting up a DHCP Server

- **Description**: Install and configure a DHCP server (e.g., using isc-dhcp-server). Set up a range of IP addresses to be dynamically assigned to devices in a local network. Test the configuration by connecting devices and verifying IP assignment.

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3. Setting up a Web Server

- **Description**: Configure a web server using Apache or Nginx. Set up a basic website and enable SSL/TLS for secure access. Configure permissions for the web content directory and explore logging and traffic monitoring features.

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4. Setting up a DNS Server

- **Description**: Install and configure a DNS server using BIND. Set up DNS zone files for a local domain and configure both forward and reverse lookup zones. Test domain name resolution and implement basic DNS security measures.

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5. Setting up a Print Server

- **Description**: Set up a print server using CUPS (Common Unix Printing System). Connect and configure a local printer to be shared over a network. Ensure that multiple users on different devices can access the printer, and set user permissions.

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6. Setting up a NFS (Network File System) Server

- **Description**: Install and configure an NFS server. Share a directory over the network and ensure proper permissions for clients to access the files. Test file sharing between multiple devices, and ensure proper security using firewall rules.

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7. Setting up a Samba Server

- **Description**: Configure a Samba server for sharing files between Linux and Windows systems. Set up user accounts and permissions for accessing shared directories. Test the connection from both Linux and Windows clients.

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8. Setting up a Proxy Server

- **Description**: Set up a proxy server using Squid or Privoxy. Configure caching to improve network performance and implement access controls to restrict usage. Test by routing network traffic through the proxy and analyzing log files.

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9. Shell Script based Project

Description: Write shell scripts that automate various system administration tasks. These tasks include process management, disk monitoring, backup automation, system health monitoring, and user management. The complexity of the project can vary based on the tasks performed, ranging from simple file handling to more advanced process and network management.

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10. Operating System Security/Protection Project

Description: Choose a security mechanism to implement, such as encryption (e.g., AES for file security) or access control mechanisms (role-based or discretionary access control). Ensure that the system can withstand common security threats (e.g., brute-force attacks, unauthorized access).

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11. Develop a Shell or Command Line Interface

Description: Develop a basic command-line interface using C or Python. Implement commands like listing files, navigating directories, and handling simple processes. Add support for command chaining (using pipes) and background process handling.